

Cracking Wordle: How Information Theory Wins 99% of the Time

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Wordle players, take note: a mathematical approach developed at Binghamton University can solve the daily puzzle with a staggering 99% success rate. The strategy, published in the Northeast Journal of Complex Systems, harnesses **Shannon entropy** a concept from information theory that measures uncertainty. Instead of guessing words with common letters like "ARISE," the method selects guesses that maximize information gain, effectively splitting the remaining possibilities into the smallest possible groups. "A subtle but important insight is that a guess doesn't have to be the most likely answer; it simply has to be informative," explains Donald Stephens, a doctoral student and co-author. The approach can seem counterintuitive because it sometimes suggests obscure words. But by focusing on reducing uncertainty rather than directly hitting the target, the solver often reaches the answer in fewer attempts. To use it, a player would run a companion script, inputting the color-coded feedback after each guess. The program then recommends the next word expected to provide the most useful information. The project began as a classroom assignment in Assistant Professor Congyu Wu's course. He challenged students to apply information theory to a real-world problem. "What is especially creative is that it transformed a static measurement into a dynamic solution," Wu says. The team compared their entropy-based method with a traditional strategy that prioritizes frequent letters. The entropy approach consistently outperformed it, solving puzzles more efficiently and rarely failing. This work not only offers a winning Wordle tactic but also demonstrates how information theory can optimize practical tasks. It's a powerful reminder that sometimes the best move isn't the most obvious one it's the one that teaches you the most.

Word Treasure

Shannon entropy -- A measure from information theory that tells how uncertain or unpredictable something is.

information theory -- A branch of mathematics that studies how data can be stored, sent, and received efficiently.

counterintuitive -- Seeming opposite to what common sense would suggest.

obscure -- Not well known; rarely used or difficult to understand.

entropy -- A measure of disorder or randomness, often used in science to talk about energy or information.

dynamic solution -- A method that changes and adapts based on new information instead of staying the same.

Background you need

Wordle, the daily word-guessing game created by Welsh software engineer Josh Wardle, became a global sensation in late 2021 and was later acquired by The New York Times for over \$1 million.

Claude Shannon, an American mathematician and engineer, founded information theory in 1948 with his landmark paper 'A Mathematical Theory of Communication,' introducing entropy as a measure of uncertainty.

Since then, entropy-based strategies have been used to solve puzzles like Mastermind and even to model decision-making in economics and biology.

The Northeast Journal of Complex Systems is a peer-reviewed academic journal that publishes interdisciplinary research on complex systems, while Binghamton University, part of the State University of New York (SUNY) system, is known for its strong computer science and systems science programs.

The Wordle-solving approach described by Donald Stephens and Congyu Wu shows how a high-level mathematical concept can be turned into a practical tool for a casual game, highlighting the power of applied mathematics in everyday life.

Break it down

LEAD: Opening hook--Wordle players can use a mathematical approach with 99% success

+ METHOD: Uses Shannon entropy from information theory to measure uncertainty

CORE INSIGHT: Guess doesn't have to be likely, just informative

+ QUOTE: Donald Stephens explains counterintuitive nature

+ EXAMPLE: May suggest obscure words to maximize information gain

PRACTICAL IMPLEMENTATION: A companion script that takes color feedback and suggests next word

+ ORIGIN: Classroom assignment in Congyu Wu's course at Binghamton University

COMPARISON: Entropy method tested against traditional frequent-letters strategy

+ RESULT: Entropy method solved puzzles more efficiently and rarely failed

TAKEAWAY: Information theory can optimize practical tasks beyond games

+ MEMORABLE: Sometimes the best move is the one that teaches you the most

Why it matters

This work demonstrates that abstract math like information theory can be applied to fun everyday puzzles, encouraging students to see the value in learning STEM concepts and potentially using them to solve larger real-world problems.

Test what you remember

1. What mathematical concept is central to the Wordle-solving strategy?
 - (A) Newton's laws
 - (B) Shannon entropy
 - (C) Pythagorean theorem
 - (D) Fibonacci sequence
2. According to the method, what should a guess prioritize?
 - (A) Using the most common letters
 - (B) Matching the last answer quickly
 - (C) Maximizing information gain
 - (D) Avoiding repeated letters
3. Where was this entropy-based Wordle strategy developed?
 - (A) Stanford University
 - (B) Binghamton University
 - (C) University of Oxford
 - (D) MIT
4. Why might the method suggest obscure words as guesses?
 - (A) Because they are easier to remember
 - (B) Because they eliminate more possibilities
 - (C) Because they use rarer letters
 - (D) Because the script has a limited dictionary
5. How does the player interact with the strategy according to the article?
 - (A) By typing all possible words
 - (B) By running a companion script and inputting color-coded feedback
 - (C) By drawing letters on paper
 - (D) By shouting guesses to a smart speaker
6. What was the origin of the project?
 - (A) A PhD thesis
 - (B) A corporate competition
 - (C) A classroom assignment by Professor Congyu Wu
 - (D) A solo hobby project by Donald Stephens

Answer key (try the quiz first!): 1: B 2: C 3: B 4: B 5: B 6: C

- 1. Power of applied mathematics:** The entropy strategy shows how a seemingly abstract theory can be turned into a near-perfect solving tool, proving that math isn't just for textbooks--it has real, fun applications. It also underscores the value of student-driven research projects.
- 2. Loss of fun and human skill:** Using an algorithm takes the human challenge out of Wordle; the joy of deducing words through intuition and vocabulary is replaced by a script's cold logic. Some argue that the game's simplicity is what made it popular, and automation could reduce its appeal.
- 3. Optimal guessing strategy:** The method reveals a deeper truth about problem-solving: the most useful guess is not necessarily the most likely answer, but the one that eliminates the most possibilities. This principle is used in many areas, from medical diagnosis to search engines.
- 4. Broader applications:** The entropy approach could be adapted to other word games, such as crossword puzzles or even spelling correction in apps. More importantly, it could inspire similar algorithms for decision-making under uncertainty, like financial forecasting or climate modeling.

MY THOUGHTS (at least 20 words):